

DNS Troubleshooting and Resolution Runbook

DNS (Domain Name System) is responsible for translating human-readable domain names into IP addresses that computers use for communication. When users report that a website is unavailable, email delivery has failed, API requests cannot reach a target service, or applications cannot connect to remote systems, DNS should always be one of the first layers investigated. A large percentage of application connectivity issues originate from DNS misconfigurations, missing records, propagation delays, nameserver failures, or resolver-related problems.

The primary objective of DNS troubleshooting is to determine whether a domain can be resolved correctly, whether the returned records are accurate, and whether the authoritative nameservers are functioning properly.

The `dns_lookup` tool should be used whenever there is a need to verify DNS records. The tool can retrieve A, AAAA, CNAME, MX, NS, TXT, PTR, and other DNS record types. The `get_dns_config` tool should be used whenever local resolver configuration must be inspected. The `retrieve_dns_context` tool should be used to access this knowledge base and retrieve documented troubleshooting procedures, historical incident patterns, and remediation guidance.

Common DNS Record Types

A Record

The A record maps a hostname to an IPv4 address.

Example:

example.com → 93.184.216.34

Common issues include:

- Incorrect IP address
- Missing record
- Stale record after migration
- Load balancer misconfiguration

Symptoms:

- Website unavailable
- Application timeout
- Users reaching old infrastructure

Recommended actions:

- Verify expected IP address
 - Validate authoritative DNS data
 - Confirm hosting migration status
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AAAA Record

The AAAA record maps a hostname to an IPv6 address.

Example:

example.com → 2606:2800:220:1:248:1893:25c8:1946

Common issues:

- Broken IPv6 routing
- Incorrect IPv6 allocation
- Partial IPv6 deployment

Symptoms:

- Website works on some networks only
- Slow connection establishment
- Intermittent failures

Recommended actions:

- Test IPv4 and IPv6 separately
 - Remove invalid AAAA records
 - Verify IPv6 connectivity
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CNAME Record

The CNAME record creates an alias to another hostname.

Example:

www.example.com → example.com

Common issues:

- Alias chain too long
- Target hostname missing
- Circular references

Symptoms:

- Resolution failures
- Increased DNS lookup latency

Recommended actions:

- Validate target hostname
 - Eliminate circular dependencies
 - Reduce alias depth
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MX Record

The MX record identifies mail servers for a domain.

Example:

example.com MX 10 mail.example.com

Common issues:

- Missing MX records
- Incorrect priorities
- Mail server hostname not resolvable

Symptoms:

- Email delivery failures
- Delayed email processing
- SMTP connection failures

Recommended actions:

- Validate mail server hostnames
- Verify priority ordering

- Confirm destination host availability
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NS Record

The NS record identifies authoritative nameservers.

Example:

example.com NS ns1.provider.net

example.com NS ns2.provider.net

Common issues:

- Invalid delegation
- Missing nameserver entries
- Nameserver mismatch

Symptoms:

- SERVFAIL responses
- Inconsistent DNS resolution
- Long propagation periods

Recommended actions:

- Compare registry delegation with authoritative configuration
 - Verify all nameservers respond consistently
 - Confirm glue record accuracy
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TXT Record

TXT records store arbitrary text information.

Common uses:

- SPF
- DKIM
- DMARC
- Domain verification

Common issues:

- Syntax errors

- Multiple conflicting SPF records
- Truncated values

Symptoms:

- Email authentication failures
- Third-party verification failures

Recommended actions:

- Validate record syntax
 - Consolidate duplicate SPF entries
 - Confirm propagation
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DNS Failure Patterns

NXDOMAIN

Meaning:

Non-Existent Domain

The domain or hostname does not exist.

Possible causes:

- Typographical error
- Deleted record
- Expired domain registration
- Incorrect zone configuration

Evidence:

Status: NXDOMAIN

Recommended actions:

1. Verify spelling.
 2. Check domain registration.
 3. Confirm record exists on authoritative servers.
 4. Review recent DNS changes.
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SERVFAIL

Meaning:

Server Failure

The DNS server could not successfully complete the request.

Possible causes:

- DNSSEC issues
- Broken delegation
- Failed authoritative nameserver
- Resolver misconfiguration

Evidence:

Status: SERVFAIL

Recommended actions:

1. Verify DNSSEC signatures.
 2. Test authoritative nameservers directly.
 3. Check nameserver availability.
 4. Validate delegation chain.
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REFUSED

Meaning:

Query refused by nameserver.

Possible causes:

- Access restrictions
- Resolver policy controls
- Misconfigured ACLs

Evidence:

Status: REFUSED

Recommended actions:

1. Verify query source.
 2. Review nameserver ACLs.
 3. Confirm intended access policy.
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Timeout

Meaning:

DNS query received no response.

Possible causes:

- Network outage
- Firewall blocking DNS
- Nameserver unavailable

Evidence:

Request timed out.

Recommended actions:

1. Check network connectivity.
 2. Verify UDP and TCP port 53 access.
 3. Confirm nameserver health.
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DNS Diagnostic Workflow

When troubleshooting DNS-related incidents, follow this sequence:

Step 1

Run:

dns_lookup

Collect:

- A records
 - AAAA records
 - NS records
 - MX records
 - CNAME records
 - TXT records
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Step 2

Evaluate response status.

Possible outcomes:

NOERROR

NXDOMAIN

SERVFAIL

REFUSED

TIMEOUT

Step 3

Validate returned records.

Check:

- Expected IP addresses
 - Correct aliases
 - Valid nameservers
 - Proper mail routing
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Step 4

Inspect resolver configuration.

Run:

get_dns_config

Review:

- Primary resolver
 - Secondary resolver
 - Search domains
 - DNS suffix configuration
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Step 5

Determine root cause category.

Common categories:

Resolver Issue

Authoritative DNS Issue

DNSSEC Issue

Delegation Issue

Zone Configuration Issue

Propagation Delay

Domain Registration Issue

Step 6

Provide evidence-based remediation.

The remediation must be supported by:

- dns_lookup results
- get_dns_config results
- Retrieved knowledge base context

Conclusions must never be based on assumptions.

If evidence is insufficient, explicitly state:

Insufficient evidence to determine root cause.

Additional DNS data is required.

Known Production Incident Patterns

Pattern: Website Unreachable but DNS Works

Evidence:

A record resolves correctly.

Ping succeeds.

HTTP connection fails.

Root Cause:

Application or service layer failure.

Recommendation:

Escalate to service diagnostics.

Pattern: Website Unreachable and NXDOMAIN Returned

Evidence:

DNS lookup returns NXDOMAIN.

Root Cause:

Missing or deleted DNS record.

Recommendation:

Restore DNS record in authoritative zone.

Pattern: Email Not Delivered

Evidence:

Missing MX records.

Root Cause:

Mail routing configuration absent.

Recommendation:

Configure valid MX records.

Pattern: Intermittent Resolution

Evidence:

One nameserver returns correct data.

Another returns outdated data.

Root Cause:

Zone replication inconsistency.

Recommendation:

Synchronize authoritative nameservers.
